Software Requirements Specification (SRS)

Unified Room Allocation and Lecture Hall Complex Management System

Version: 2.3
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Source Code Repository: Github Link
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Revision History

Revision	Date	Reason for Change	Proposed By	Worked On By	Version
Eight Revision	14 Feb	Modified the following UML diagrams: Class, activity, use case. Created the following new diagrams: timing, component.	Team	Team	2.4
Seventh Revision	4 Feb	Updated UML class diagrams to include more detailed description of attributes; UML use case diagram to have more emphasis on user/actor interaction; added more details in other diagrams	Team	Team	2.3
Sixth Revision	4 Feb	Revised booking conflict handling to replace peer negotiation with admin-driven resolution and system notifications; added section 4.3.6 containing REQ-4.3.14	Team	Team	2.2

Fifth revision	4 Feb 2026	Updated the RTM to reflect changes made in fourth revision	Team	Team	2.1
Fourth Revision	30 Jan 2026	Revised user access model to allow student-initiated room booking requests with mandatory faculty verification prior to LHC staff approval. (In 4.1.2 and REQ-4.1.5)	Team	Team	2.0
Third Revision	28 Jan 2026	Added requirement priorities, formal NFR template, explicit traceability matrix	Team/Reviewer Feedback	Team	1.3
Second Revision	21 Jan 2026	Added functional requirements regarding slot and class venue change (Section 4.3)	Team	Team	1.2
First Revision	20 Jan 2026	Added functional requirements (REQ-4.1.4 to REQ-4.1.9)	Team	Team	1.1
Initial Version	19 Jan 2026	Initial SRS document	Team	Team	1.0

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines the functional and non-functional requirements for the Unified Room Allocation and Lecture Hall Complex (LHC) Management System. The system aims to automate room booking, timetable-based classroom allocation, slot management, and examination seating arrangements in a college environment. This document serves as a reference for developers, testers, project managers, and stakeholders.

1.2 Document Conventions

- Functional requirements are labeled as REQ-x.y.z (Priority).
- Non-functional requirements are labeled as NFR-x.y.z (Priority).
- “Shall” indicates a mandatory requirement.

- “Should” indicates a desirable but non-mandatory feature.
- Priority levels: High / Mid / Low.
- TBD denotes requirements to be finalized later.

1.3 Intended Audience and Reading Suggestions

Developers, testers, administrators, faculty representatives, and student representatives. Readers new to the project should begin with Sections 1 and 2, followed by Sections 4 and 5.

1.4 Product Scope

The product is a web-based portal providing a unified gateway for managing lecture halls, department rooms, events, and examination schedules. It eliminates manual spreadsheet-based booking, prevents clashes, improves transparency, and enables intelligent scheduling and recommendations.

1.5 References

- IEEE Software Requirements Specification Template (Karl E. Wiegers)
- Project Description Document (SE Documentation)

1.6 Alpha Managers

Name	**Alpha Managers**
Aryan Ashok Jain	Requirements Software System
Suparn Agrawal	Requirements Work Opportunity
Rewant Rai	Requirements StakeHolders Way of Working
Anshika Jha	Requirements Team

2. Overall Description

2.1 Product Perspective

The system is a new, centralized web application replacing the current manual email-and-spreadsheet-based LHC booking process. It integrates timetable data, room inventories, user roles, and scheduling algorithms into a single platform. It may later integrate with existing academic databases (student/course information).

2.2 Product Functions

At a high level, the system will:

- Maintain real-time room availability status.
- Pre-book classrooms based on released slot-based timetables.
- Allow users to request; and LHC staff to approve, or reject room bookings.
- Handle slot changes and room change requests.
- Generate examination timetables and seating plans.
- Provide dashboards tailored to different user roles.
- Send notifications and reminders.

2.3 User Classes and Characteristics

Admin
Uploads finalized timetables, academic calendars, institute holidays, examination schedules, and global system constraints.

LHC Staff

- High system privileges.
- Manages rooms, departments, and allocation policies.
- Responsible for final approval or rejection of all room booking requests.

Faculty

- Moderate technical proficiency.
- Can initiate room booking requests for course-related and miscellaneous activities.
- Acts as a verifier for student-initiated room booking requests.
- Can approve or reject student booking requests before forwarding them to LHC staff.

Students (General Access)

- All enrolled students are permitted to initiate room booking requests.
- Must specify the purpose of booking (event type, student body/society, or reason).

- Must select a faculty member (preferably the faculty advisor of the concerned activity) for verification.
- Student booking requests are forwarded to the selected faculty for verification before being sent to LHC staff.

Rationale:
This model ensures inclusive access while enforcing accountability and legitimacy through faculty verification.

2.4 Operating Environment

- Web-based application.
- Runs on modern browsers (Chrome, Firefox, Edge).
- Server-side: Linux/Windows server.
- Database: Relational DBMS (e.g., MySQL/PostgreSQL).

2.5 Design and Implementation Constraints

- Must support multiple slot systems (e.g., First Year vs Senior Years).
- Must avoid scheduling clashes for students enrolled in multiple courses.
- Role-based access control is mandatory.
- Algorithms must consider room capacity and constraints.

2.6 User Documentation

- Online user manual.
- Admin guide.
- Context-sensitive help within the portal.

2.7 Assumptions and Dependencies

- Timetables are released digitally in a structured format.
- Student-course enrollment data is accurate.
- Users have reliable internet access.

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3. External Interface Requirements

3.1 User Interfaces

- Web-based GUI with dashboards per user role.
- Visual room and seat maps (highlighted seating during exams).
- Forms for booking requests with optional messages (e.g., mic/projector).

3.2 Hardware Interfaces

- No direct hardware interfaces required.

3.3 Software Interfaces

- Database system for persistent storage.
- Optional integration with academic ERP systems.

3.4 Communications Interfaces

- Uses HTTP/HTTPS.
- Email and in-app notifications.

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4. System Features

4.1 Room Booking and Allocation

4.1.1 Description and Priority

Allows authorized users to view availability and request rooms. **Priority: High**.

4.1.2 Stimulus/Response Sequences

1. User (Faculty or Student) selects date, slot, and room.
2. User specifies booking purpose and required event parameters.
3. **If the requester is a Student:**
    - System requires selection of a faculty verifier.
    - Request is forwarded to the selected faculty for verification.

    - Faculty approves or rejects the request.
    - Approved requests are forwarded to LHC staff.
4. **If the requester is a Faculty member:**
    - Request is directly forwarded to LHC staff.
5. The system validates availability and constraints.
6. LHC staff reviews and approves or rejects the request.
7. **Conflict handling:** If multiple booking requests conflict for the same room and slot, the system escalates the conflict to the Admin for resolution and notifies all affected users of the final decision.

4.1.3 Functional Requirements

- **REQ-4.1.1 (High):** System *shall* display real-time room availability.
- **REQ-4.1.2 (High):** System *shall* prevent double booking.
- **REQ-4.1.3 (High):** System *shall* notify staff of new requests.
- **REQ-4.1.4 (High):** System *shall* provide an option for authorized staff to upload the complete semester timetable at once in the form of an Excel sheet at the start of the semester, parse the uploaded data, and automatically reflect all class-slot-room allocations in the system database.
- **REQ-4.1.5 (High):** System shall allow faculty and students to initiate room booking requests for specific events or purposes.
- **REQ-4.1.6 (Mid):** The system *should* have options for various kinds of event types for room booking (e.g., quiz, speaker session, seminar, meeting).
- **REQ-4.1.7 (Mid):** System shall require users to provide event-specific parameters based on the selected event type, such as number of participants, seating constraints, invited guest details, and special requirements.
- **REQ-4.1.8 (High):** System shall validate a booking request against potential clashes, including but not limited to:
    - room unavailability <High>,
    - insufficient room capacity <High>,
    - conflicting bookings <High>, and
    - invited guests' existing commitments <Mid>.
- **REQ-4.1.9 (High):** If any clash is detected, the system shall prevent the request from being raised and
    - *should* suggest alternative rooms, slots, or configurations that satisfy the given constraints <Mid>.
- **REQ-4.1.10 (Low):** In case of conflicting booking requests for the same room and slot, the system shall forward reconsideration requests to the Admin, who shall decide which request is approved. Both users shall be notified of the decision.

4.2 Timetable-Based Preallocation

Description and Priority

Automatically allocates classrooms based on slot timetables.

- **REQ-4.2.1 (High):** System shall ingest timetable data.
- **REQ-4.2.2 (Mid):** System shall map courses to rooms based on capacity.

4.3 Slot and Class Venue Exchange Facilitation and Some other Extensional Features (Optional Extension – Phase 2)

4.3.1 Description and Priority

This feature enables faculty to request changes to the allocated time slot of a course and facilitates intelligent evaluation of feasibility. The system validates constraints related to students, room availability, and room capacity before allowing a slot change request to proceed.

4.3.2 Stimulus/Response Sequences (Slot Change)

- Course instructor selects an existing course offering.
- Instructor requests to move the course from Slot A to Slot B.
- System evaluates feasibility by checking:
    - Availability of Slot B.
    - Availability of an appropriate classroom during Slot B.
    - Potential clashes for students enrolled in the course (other classes or commitments).
- If no limiting constraints exist:
    - System allows the instructor to proceed with the slot change request.
    - LHC staff are notified for review and final approval.
- If constraints exist:
    - System blocks the slot change request.
    - System displays the reasons for rejection.
    - System should recommend alternative feasible slots and/or rooms.

4.3.3 Functional Requirements (Slot Change)

- **REQ-4.3.1 (Mid):** System shall allow a course instructor to request a change of the course time slot from an existing slot to a desired target slot.
- **REQ-4.3.2 (Mid):** System shall validate the requested target slot against student enrollment data to detect any student-level schedule conflicts.
- **REQ-4.3.3 (Mid):** System shall validate the availability of at least one classroom of appropriate capacity during the requested target slot.
- **REQ-4.3.4 (Mid):** If no limiting constraints are detected, the system shall allow the slot change request to be raised and shall notify the LHC staff for approval.
- **REQ-4.3.5 (Mid):** If one or more limiting constraints are detected, the system shall prevent the slot change request from being raised.
- **REQ-4.3.6 (Mid):** When a slot change request is blocked, the system shall present clear reasons for the infeasibility.
- **REQ-4.3.7 (Low):** In case of infeasibility, the system shall recommend alternative slots that satisfy student availability and classroom capacity constraints.

4.3.4 Class Venue Change Facilitation

Description and Priority
This feature allows a course instructor to request a change in the physical classroom (venue) for an already scheduled course. The system evaluates room availability across different slots and assists in identifying feasible alternatives.

Stimulus/Response Sequences (Venue Change)

- Course instructor selects an existing course and its currently allocated classroom (e.g., Room 101).
- Instructor requests to conduct the course in a different classroom (e.g., Room 201).
- System checks the availability of the requested room across relevant slots.
- System identifies and displays slots in which the requested room is free.
- Instructor selects a suitable slot from the suggested options.
- System raises a venue change request and notifies the LHC staff for approval.

Functional Requirements (Venue Change)

- **REQ-4.3.8 (Mid):** System shall allow a course instructor to request a change of classroom for an already allocated course.
- **REQ-4.3.9 (Mid):** System shall evaluate the availability of the requested classroom across different slots.
- **REQ-4.3.10 (Mid):** System shall suggest feasible slots in which the requested classroom is available.
- **REQ-4.3.11 (Mid):** Upon instructor confirmation, the system shall raise a venue change request and notify the LHC staff for approval.

4.3.5 Metadata-Based Venue Recommendation

In Phase II, the system may maintain detailed room metadata such as availability of projectors, seating type, accessibility features, and room category.

- **REQ-4.3.12 (Optional/Mid):** System should allow instructors to specify required room features instead of a specific room.
- **REQ-4.3.13 (Optional/Mid):** Based on the specified features, the system should recommend suitable rooms along with feasible slots.

4.3.6 Dependency on Student-Enrollment Data

Student enrollment data for clash detection is expected to be provided by the Admin; however, the system shall remain operational in its absence.

- **REQ-4.3.14 (Mid):** System shall perform student-level clash validation for slot and venue change requests only if student enrollment data is available. In the absence of such data, the system shall bypass student availability checks, continue evaluating other

applicable constraints (such as room availability and capacity), and clearly inform the user that student-level validation was not performed. This is in accordance with NFR5.4.1

4.4 Examination Scheduling and Seating

- **REQ-4.4.1 (Low):** System shall generate exam timetables based on constraints.
- **REQ-4.4.2 (Low):** System shall assign seats to students without clashes.
- **REQ-4.4.3 (Low):** System shall visually display seating arrangements.

4.5 Notifications and Messaging

- **REQ-4.5.1 (Mid):**
  System shall send reminders and status updates for booking requests.
- **REQ-4.5.2 (Mid):**
  System shall notify faculty members of pending student booking verification requests.
- **REQ-4.5.3 (Mid):**
  System shall notify students when their booking requests are approved, rejected, or forwarded to LHC staff.

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5. Non-Functional Requirements

5.0 Non-Functional Requirement Template

Each non-functional requirement is specified using the following structure:

- **Description** – What quality aspect is being addressed.
- **Measurable Criterion** – Quantitative or verifiable condition.
- **Method of Verification** – How the requirement will be tested or validated.
- **Priority** – High / Mid / Low.
- **Affected Modules** – Primary system components involved.

5.1 Performance Requirements

NFR-5.1.1 (High)

- Description: System responsiveness for room availability queries.

- Measurable Criterion: 95% of room availability queries shall respond within 2 seconds.
- Method of Verification: Load and performance testing.
- Affected Modules: Booking module, Database, Availability service.

5.2 Safety Requirements

NFR-5.2.1 (High)

- Description: Protection against accidental data loss.
- Measurable Criterion: Critical administrative data shall not be permanently deleted without multi-step confirmation and backup logging.
- Method of Verification: Fault-injection testing, audit review.
- Affected Modules: Admin console, Database layer.

5.3 Security Requirements

NFR-5.3.1 (High)

- Description: Role-based system security.
- Measurable Criterion: All protected operations must require authenticated and authorized access based on role definitions.
- Method of Verification: Penetration testing, access-control testing.
- Affected Modules: Authentication service, Authorization middleware.

NFR-5.3.2 (High)

- Description: Secure handling of academic and personal data.
- Measurable Criterion: All sensitive data must be encrypted in transit and at rest.
- Method of Verification: Security audit, encryption validation.
- Affected Modules: Database, Communication layer.

5.4 Software Quality Attributes – Robustness

NFR-5.4.1 (Mid)

- Description: Graceful handling of missing administrative data.
- Measurable Criterion: System continues operation with partial validation where data is unavailable.

- Method of Verification: Negative testing.
- Affected Modules: Validation engine, Booking workflow.

NFR-5.4.2 (Mid)

- Description: Selective bypass of unavailable validations.
- Measurable Criterion: Missing datasets do not cause full process failure.
- Method of Verification: Integration testing.
- Affected Modules: Slot change, Clash detection.

NFR-5.4.3 (Mid)

- Description: User awareness of skipped validations.
- Measurable Criterion: System displays clear notices when checks are bypassed.
- Method of Verification: UI and functional testing.
- Affected Modules: UI layer, Notification service.

NFR-5.4.4 (High)

- Description: Operational stability.
- Measurable Criterion: No crashes or inconsistent system states due to missing data.
- Method of Verification: Stress testing, fault injection.
- Affected Modules: Core services, Database layer.

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6. Other Requirements

- Configurable holidays.
- Support for adding/removing departments and rooms.

Appendix A: Glossary

- **LHC:** Lecture Hall Complex.
- **Slot:** Predefined time block.
- **Staff**: Personnel for handling management of particular building
- **Admin:** Stands for relevant administrative institute authority
- **POR:** Position of Responsibility

Appendix B: Analysis Models

- Use case diagrams

- Activity diagrams
- Class diagrams
- Sequence diagrams

Appendix C: To Be Determined List

1. A Final exam scheduling algorithm.
2. ERP integration details.

Appendix D: Requirements Traceability Matrix (RTM)

Requirement ID	Source / System Goal	Related Use Case	Planned Module	Related UML Artifact	Verification Artifact
REQ-4.1.1	Real-time availability	UC-RoomBooking	Availability Service	Usecase: View Room Availability Activity: Availability check decision Sequence: BookingService -> RoomService Class: Room, Slot, Booking	TC-B01
REQ-4.1.5	Inclusive booking with verification	UC-StudentRoomBooking	Booking workflow	Use Case: Request Booking, Verify Student Request Activity: Student -> Faculty -> Staff flow Timing: Pending_Faculty -> Pending_Staff Class: BookingRequest, Student, Faculty, Staff	TC-SB01
REQ-4.1.8	Clash prevention	UC-RoomBooking	Clash Detection Engine	Use Case: Validate Slot, Check Availability Activity: Validation & alternative loop Sequence: Pre-creation validation flow	TC-B05

				Class: Slot, Room, Enrollment, BookingRequest	
REQ-4.2.1	Automated allocation	UC-Prealloc ation	Timetable Ingestor	Use Case: Upload Semester Timetable Sequence: Timetable ingestion Class: TimetableEntry, Course, Room	TC-T02
REQ-4.3.2	Slot change safety	UC-SlotCha nge	Slot Manager	Use Case: Request Slot Change Activity: Slot validation branch Sequence: Slot validation logic Class: SlotChangeRequest, Slot, Enrollment	TC-S03
REQ-4.3.8	Venue change	UC-VenueC hange	Room Manager	Use Case: Request Venue Change Activity: Venue validation flow Class: VenueChangeRequest, Room, Booking	TC-V01
REQ-4.5.1	Communic ation	UC-Notificati ons	Notificatio n Service	Use Case: Send Notification Activity: Approval/Rejection notification Sequence: NotificationService interaction Class: Notification, User	TC-N01
NFR-5.1.1	Performanc e	All major flows	API Gateway	Sequence: Layered service interaction Composite: BookingService, RoomService	Load-T est-01

				Deployment: App Server, DB Server	
NFR-5.3.1	Security	All secured cases	Auth Service	Use Case: Role-based access Class: User inheritance (Admin, Staff, Faculty, Student) Deployment: HTTPS+Auth layer	Sec-Audit-01
REQ-4.3.14	Conditional validation	UC-SlotChange	Validation Engine	Activity: Conditional validation branch Sequence: Optional enrollment check Class: Enrollment	TC-S04
REQ-4.1.10	Conflict resolution	UC-Conflict Resolution	Admin Conflict Manager	Use Case: ConflictResolution Activity: Escalation branch Class: BookingRequest, Admin	TC-CR 01

Appendix E: Use Case Specifications

This appendix formally defines the use cases referenced throughout the SRS and the Requirements Traceability Matrix (RTM). Each use case represents a high-level user–system interaction that realizes one or more functional requirements.

UC-RoomBooking

Primary Actors: Faculty, Students

Supporting Actors: Faculty (as verifier for student requests), LHC Staff, Admin

Goal:
To request and allocate a room for a specified time slot and purpose.

Preconditions:

- User is authenticated and authorized.
- Room inventory and slot information are available.

Main Flow:

1. User selects date, slot, room, and booking purpose.
2. User provides required event-specific parameters.
3. If the requester is a student, the system forwards the request to the selected faculty verifier.
4. Faculty verifies and approves the request.
5. System forwards the request to LHC staff.
6. LHC staff reviews and approves or rejects the request.
7. System updates room allocation and notifies relevant users.

Alternate Flows:

- If clashes are detected, the request is blocked or escalated for reconsideration.
- If student enrollment data is unavailable, student-level clash checks are bypassed with user notification.

Postconditions:

- Room is allocated, rejected, or escalated for administrative resolution.

UC-Preallocation

Primary Actor: Admin

Supporting Actors: System

Goal:
To automatically allocate classrooms based on released slot-based timetables.

Preconditions:

- Admin has uploaded a valid timetable file.
- Room inventory data is available.

Main Flow:

1. Admin uploads timetable data.
2. System parses timetable information.
3. System maps courses to rooms based on capacity and constraints.

4. Allocations are stored and made visible to users.

Postconditions:

- Class-slot-room mappings are established in the system.

UC-SlotChange

Primary Actor: Faculty

Supporting Actors:LHC Staff, System

Goal:
To request a change in the scheduled time slot of a course.

Preconditions:

- Course is already scheduled.
- Faculty is authorized for the course.

Main Flow:

1. Faculty selects an existing course offering.
2. Faculty requests a change from the current slot to a target slot.
3. System evaluates feasibility (room availability, capacity, student clashes).
4. If feasible, request is forwarded to LHC staff for approval.
5. LHC staff approves or rejects the request.

Alternate Flows:

- If student enrollment data is unavailable, student-level clash checks are bypassed with notification.
- If constraints fail, the system provides reasons and possible alternatives.

Postconditions:

- Course slot is updated or remains unchanged.

UC-VenueChange

Primary Actor: Faculty

Supporting Actors:LHC Staff, System

Goal:
To change the physical classroom assigned to a scheduled course.

Preconditions:

- Course is already scheduled in a room.

Main Flow:

1. Faculty selects a course and its current venue.
2. Faculty requests a different room.
3. System checks room availability across slots.
4. System suggests feasible slots or confirms availability.
5. Faculty confirms the selection.
6. Request is forwarded to LHC staff for approval.

Postconditions:

- Course venue is updated or request is rejected.

UC-Notifications

Primary Actor: System

Supporting Actors: Faculty, Students, LHC Staff, Admin

Goal:
To notify users of booking requests, approvals, rejections, and required actions.

Preconditions:

- A relevant system event has occurred.

Main Flow:

1. System detects a state change (request submission, approval, rejection, escalation).
2. System sends notifications to affected users via configured channels.

Postconditions:

- Users are informed of the current status of relevant requests.

UC-ConflictResolution *(Implicit / Administrative)*

Primary Actor: Admin

Supporting Actors: System

Goal: To resolve conflicting booking requests for the same room and slot.

Preconditions:

- Multiple booking requests conflict.

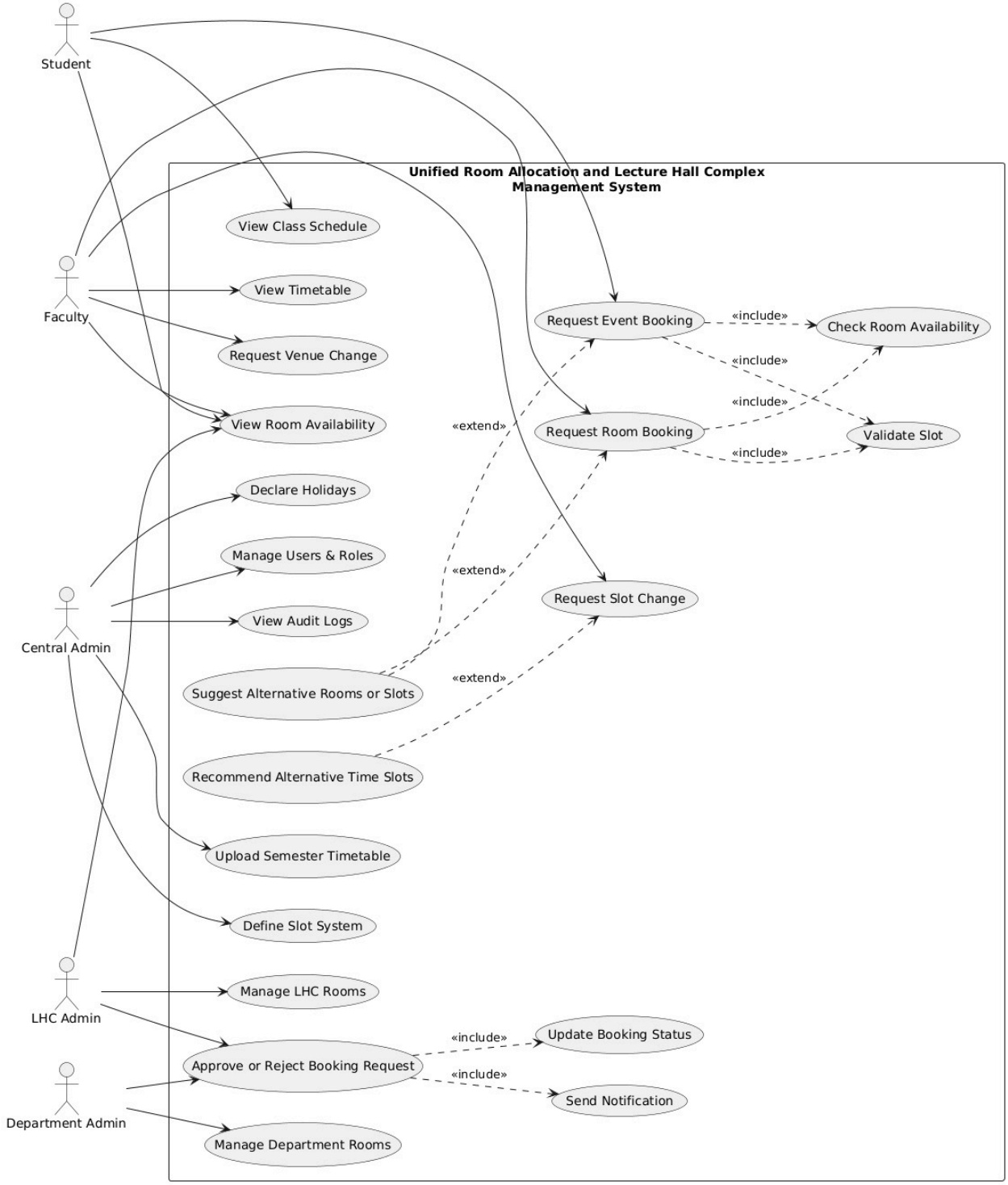
Main Flow:

1. System escalates the conflict to the Admin.
2. Admin selects one request for approval.
3. System rejects the other request(s).
4. All affected users are notified of the decision.

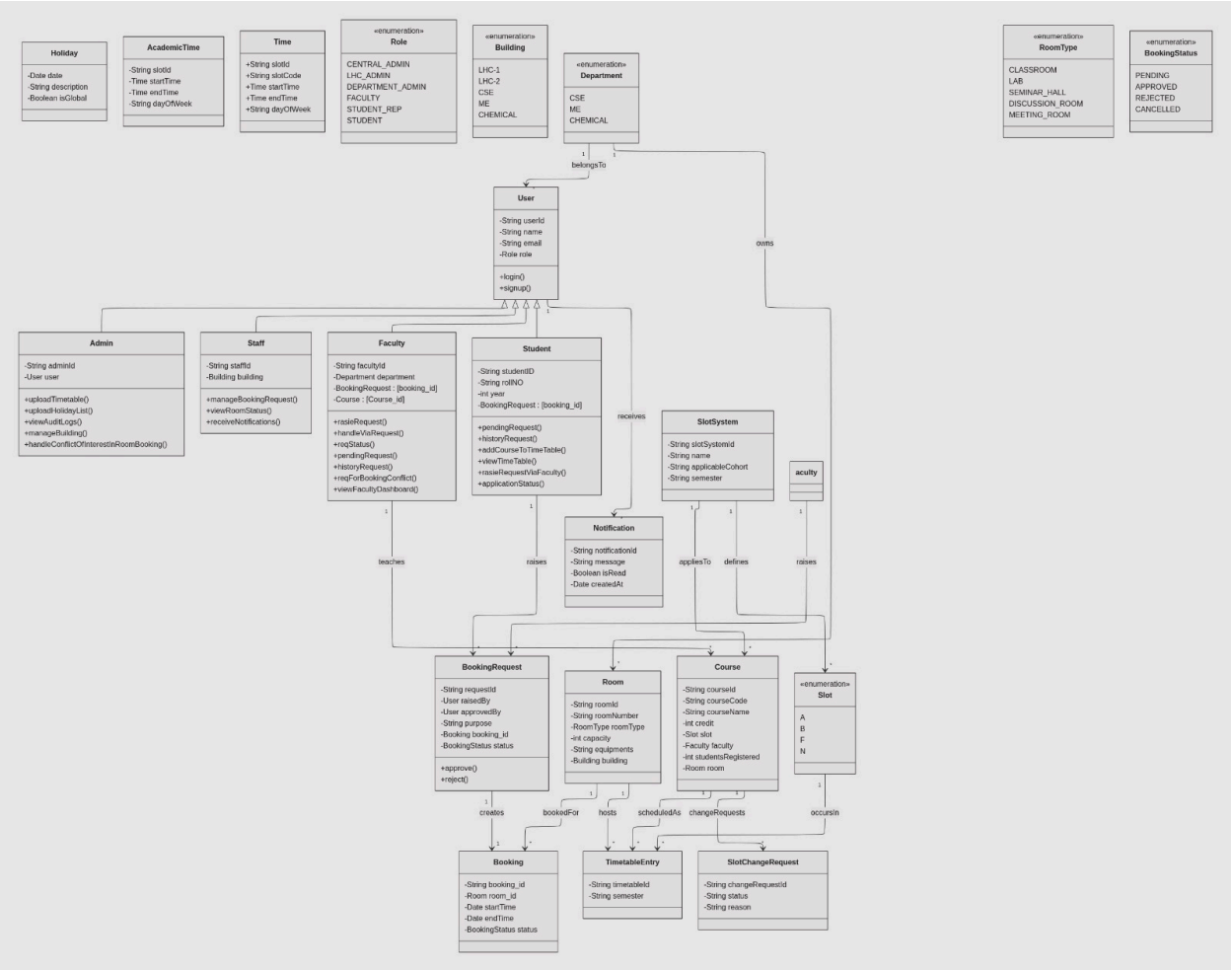
Postconditions:

- Conflict is resolved and allocation finalized.

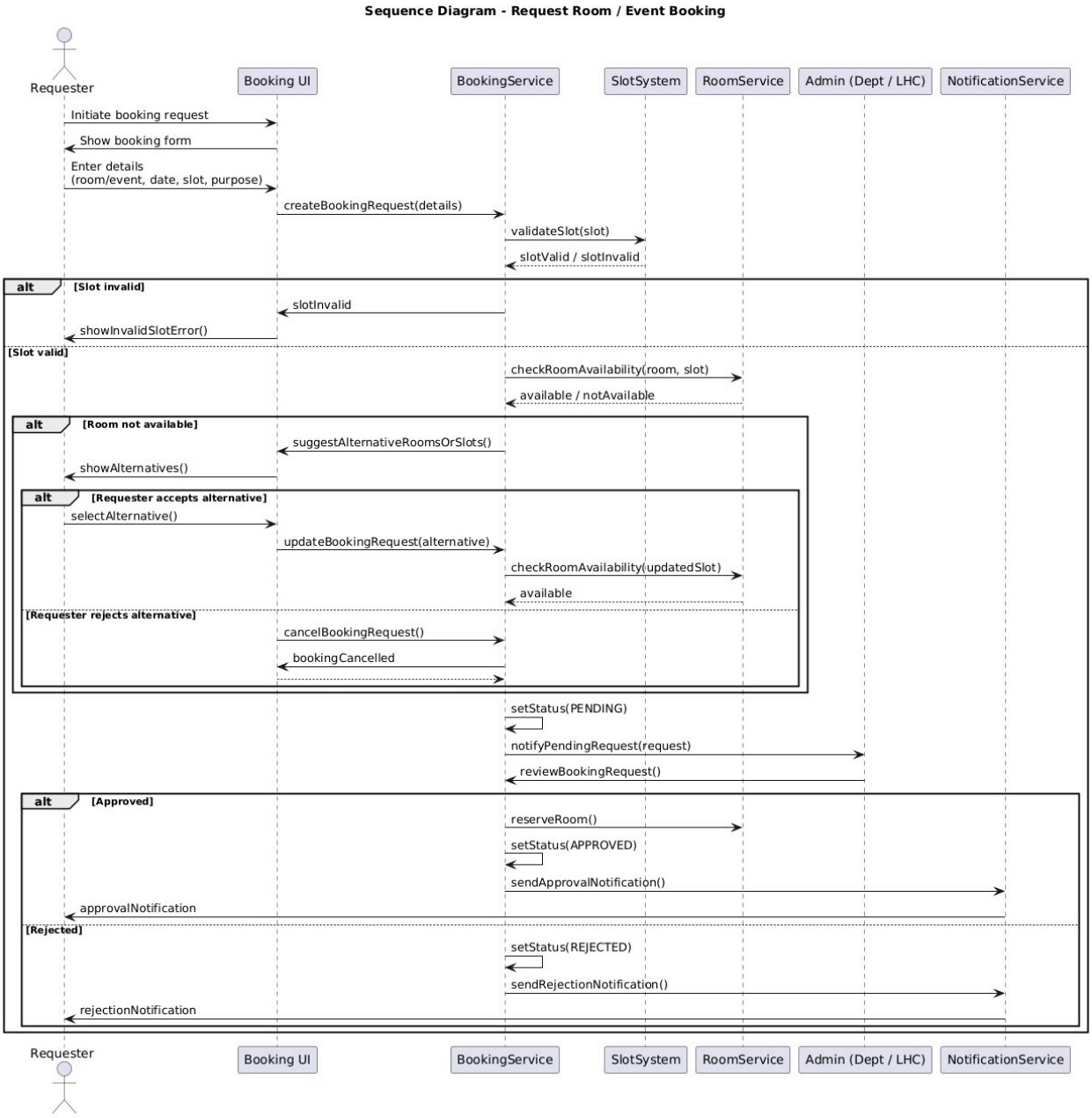
Use Case Diagram



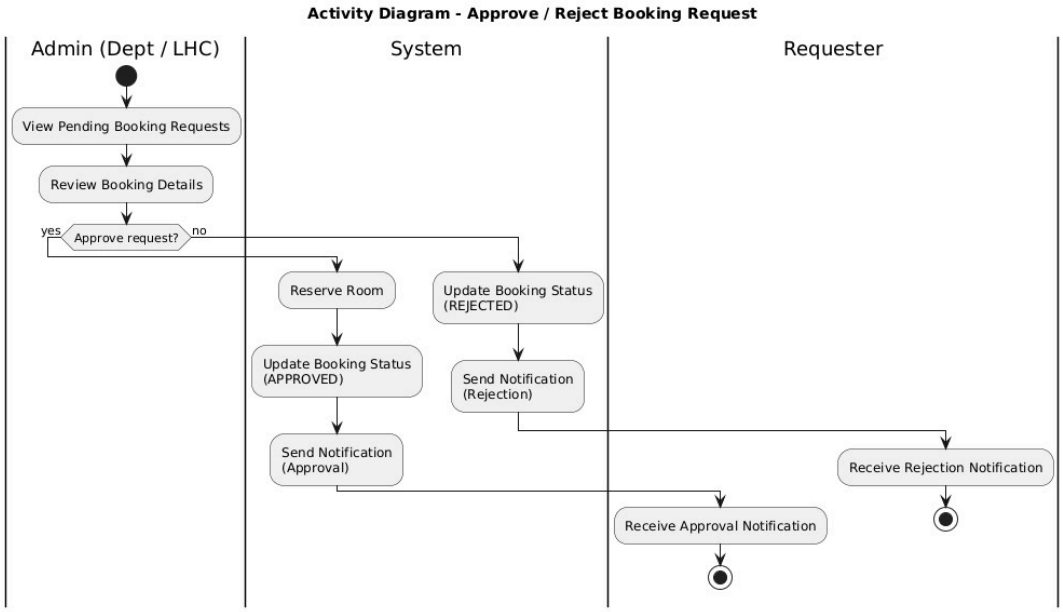
Class Diagram



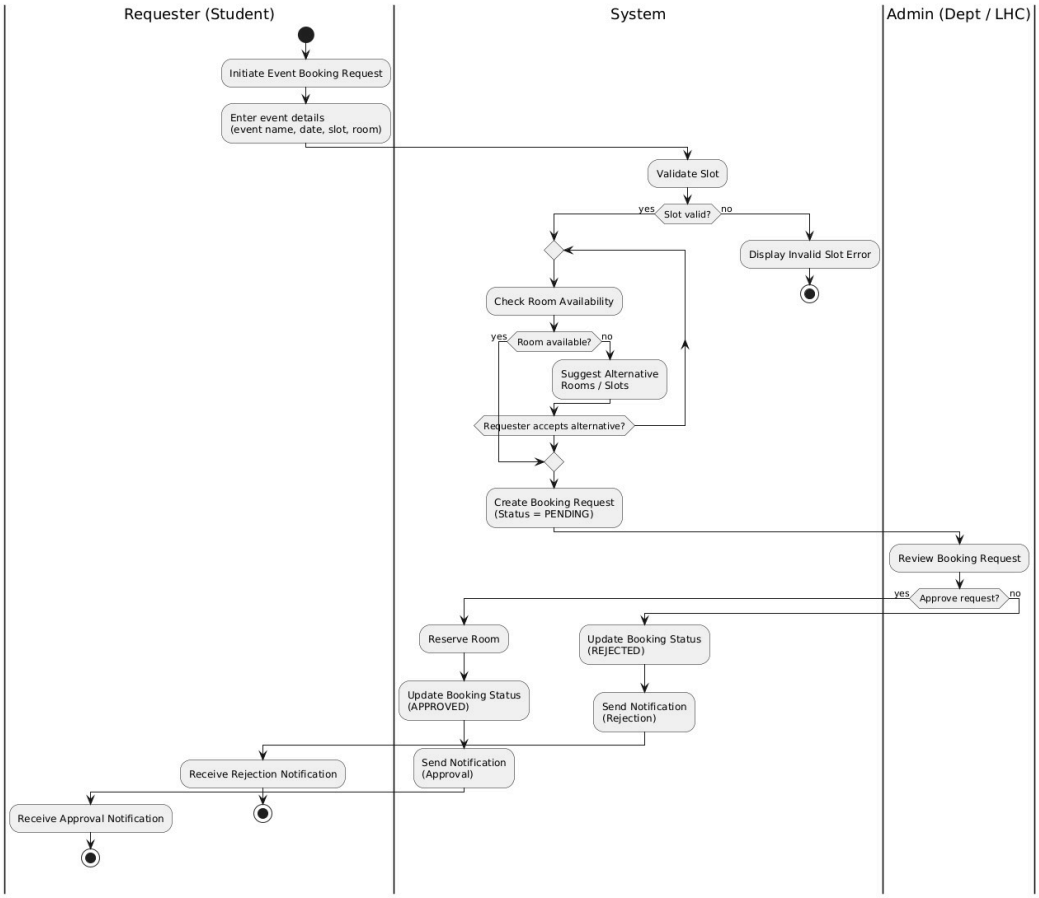
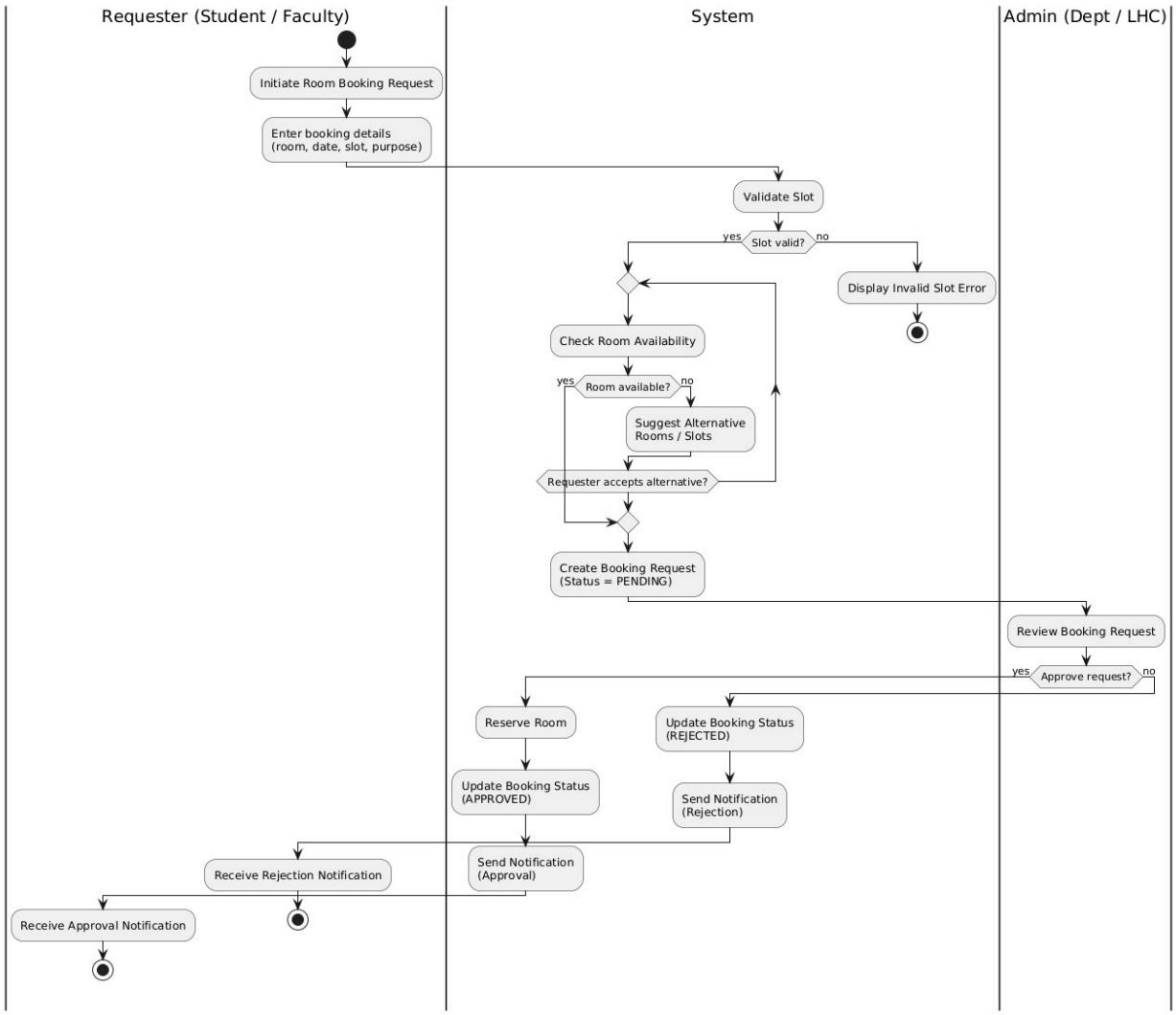
Sequence Diagram



Activity Diagram



Activity Diagram - Request Event Booking



Activity Diagram - Request Room Booking